

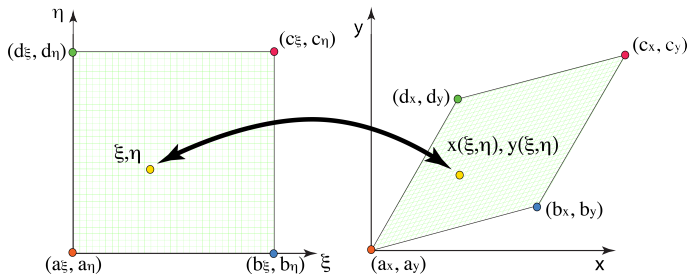
MECH5200 Numerical Methods for PDEs : *Video 12:* *2D Finite Difference Mappings – Bilinear Example*

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Idea

- Let's examine the problem of mapping from the **computational** to the **physical** domain:



- Assume we know the physical domain coordinates:
 $(a_x, a_y), (b_x, b_y), (c_x, c_y), (d_x, d_y)$
- We also define the computational domain coordinates:
 $(a_\xi, a_\eta) = (0, 0), (b_\xi, b_\eta) = (1, 0), (c_\xi, c_\eta) = (1, 1), (d_\xi, d_\eta) = (0, 1).$

Idea

- For this problem we will assume a **bilinear** mapping exists such that:

$$x(\xi, \eta) = R_x \xi \eta + S_x \xi + T_x \eta + U_x$$

$$y(\xi, \eta) = R_y \xi \eta + S_y \xi + T_y \eta + U_y$$

- We need to find the coefficient values: $R_x, R_y, S_x, S_y, T_x, T_y, U_x, U_y$.
 - We have 8-unknowns. 4 for the x -direction and 4 for the y -direction.
 - What are the known values that we can use to resolve this?

Bilinear Mapping

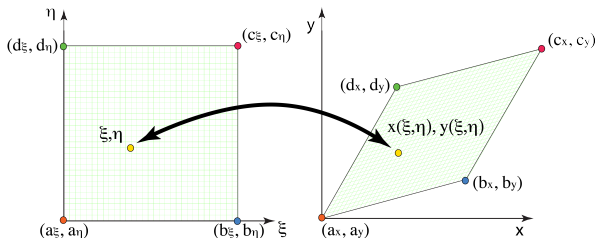
- To find $R_x, R_y, S_x, S_y, T_x, T_y, U_x, U_y$, we could plug in known coord. values of (ξ, η) to get $x(\xi, \eta)$ and $y(\xi, \eta)$, e.g.,:

$$x(\xi = 0, \eta = 0) = a_x = R_x \cdot 0 \cdot 0 + S_x \cdot 0 + T_x \cdot 0 + U_x = U_x$$

$$x(\xi = 1, \eta = 0) = b_x = R_x \cdot 1 \cdot 0 + S_x \cdot 1 + T_x \cdot 0 + U_x = S_x + U_x$$

$$x(\xi = 1, \eta = 1) = c_x = R_x \cdot 1 \cdot 1 + S_x \cdot 1 + T_x \cdot 1 + U_x = R_x + S_x + T_x + U_x$$

$$x(\xi = 0, \eta = 1) = d_x = R_x \cdot 0 \cdot 1 + S_x \cdot 0 + T_x \cdot 1 + U_x = T_x + U_x$$



Bilinear Mapping

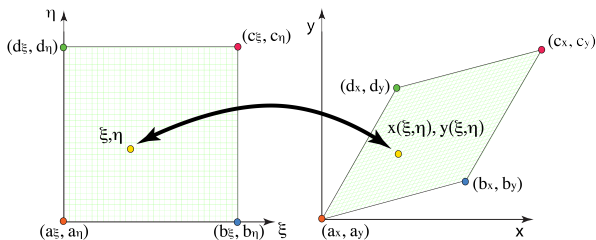
- Similarly:

$$y(\xi = 0, \eta = 0) = a_y = R_y \cdot 0 \cdot 0 + S_y \cdot 0 + T_y \cdot 0 + U_y = U_y$$

$$y(\xi = 1, \eta = 0) = b_y = R_y \cdot 1 \cdot 0 + S_y \cdot 1 + T_y \cdot 0 + U_y = S_y + U_y$$

$$y(\xi = 1, \eta = 1) = c_y = R_y \cdot 1 \cdot 1 + S_y \cdot 1 + T_y \cdot 1 + U_y = R_y + S_y + T_y + U_y$$

$$y(\xi = 0, \eta = 1) = d_y = R_y \cdot 0 \cdot 1 + S_y \cdot 0 + T_y \cdot 1 + U_y = T_y + U_y$$



Bilinear Mapping

- In the x — direction:

$$a_x = U_x$$

$$b_x = S_x + U_x$$

$$c_x = R_x + S_x + T_x + U_x$$

$$d_x = T_x + U_x$$

- In the x —direction, we can form a matrix system to solve the linear system of 4 equations:

$$\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} R_x \\ S_x \\ T_x \\ U_x \end{pmatrix} = \begin{pmatrix} a_x \\ b_x \\ c_x \\ d_x \end{pmatrix}$$

Bilinear Mapping

- Similarly, in the y – direction:

$$a_y = U_y$$

$$b_y = S_y + U_y$$

$$c_y = R_y + S_y + T_y + U_y$$

$$d_y = T_y + U_y$$

- In the y –direction, we can form a matrix system to solve the linear system of 4 equations:

$$\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} R_y \\ S_y \\ T_y \\ U_y \end{pmatrix} = \begin{pmatrix} a_y \\ b_y \\ c_y \\ d_y \end{pmatrix}$$

Bilinear Mapping

- Similar matrix in the x - and y - directions:

$$\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} R_x \text{ or } R_y \\ S_x \text{ or } S_y \\ T_x \text{ or } T_y \\ U_x \text{ or } U_y \end{pmatrix} = \begin{pmatrix} a_x \text{ or } a_y \\ b_x \text{ or } b_y \\ c_x \text{ or } c_y \\ d_x \text{ or } d_y \end{pmatrix}$$

- Solve for coefficients using `\` operator in MATLAB.

Bilinear Mapping

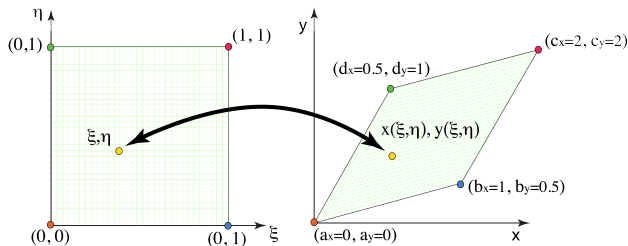
- So, with known coefficients, we can map any (ξ, η) combination to the (x, y) space:

$$x(\xi, \eta) = R_x \xi \eta + S_x \xi + T_x \eta + U_x$$

$$y(\xi, \eta) = R_y \xi \eta + S_y \xi + T_y \eta + U_y$$

Example:

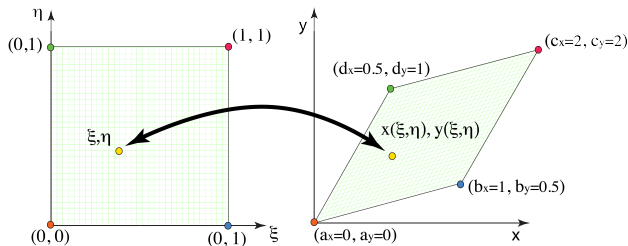
- Let's examine the following mapping from the computational to the physical domain:



$$\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} R_x \\ S_x \\ T_x \\ U_x \end{pmatrix} = \begin{pmatrix} 0.0 \\ 1.0 \\ 2.0 \\ 0.5 \end{pmatrix} \rightarrow \begin{pmatrix} R_x = 0.5 \\ S_x = 1.0 \\ T_x = 0.5 \\ U_x = 0.0 \end{pmatrix}$$

Example

- Let's examine the following mapping from the computational to the physical domain:



$$\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} R_y \\ S_y \\ T_y \\ U_y \end{pmatrix} = \begin{pmatrix} 0.0 \\ 0.5 \\ 2.0 \\ 1.0 \end{pmatrix} \rightarrow \begin{pmatrix} R_y = 0.5 \\ S_y = 0.5 \\ T_y = 1.0 \\ U_y = 0.0 \end{pmatrix}$$

Example

- The equation to map from computational domain to x -coordinate in the physical domain:

$$\begin{aligned}x(\xi, \eta) &= R_x \xi \eta + S_x \xi + T_x \eta + U_x \\x(\xi, \eta) &= 0.5 \cdot \xi \eta + 1.0 \cdot \xi + 0.5 \cdot \eta\end{aligned}$$

- The equation to map from computational domain to y -coordinate in the physical domain:

$$\begin{aligned}y(\xi, \eta) &= R_y \xi \eta + S_y \xi + T_y \eta + U_y \\y(\xi, \eta) &= 0.5 \cdot \xi \eta + 0.5 \cdot \xi + 1.0 \cdot \eta\end{aligned}$$

Example

- The equation to map from computational domain to x - and y -coordinates in the physical domain:

$$x(\xi, \eta) = 0.5 \cdot \xi\eta + 1.0 \cdot \xi + 0.5 \cdot \eta$$

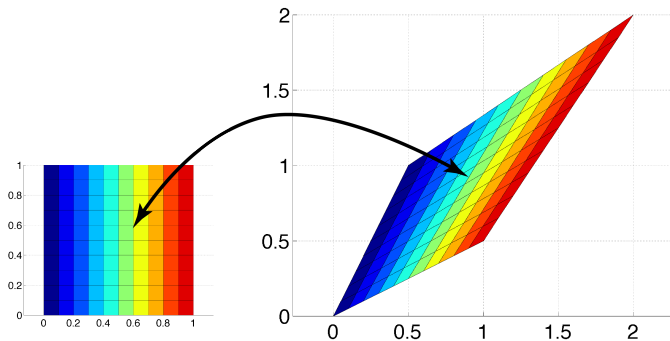
$$y(\xi, \eta) = 0.5 \cdot \xi\eta + 0.5 \cdot \xi + 1.0 \cdot \eta$$

- Let's look at this in MATLAB.
 - Cycle through a large number of points in $\xi - \eta$ computational domain.
 - Now compute the $x(\xi, \eta)$ and $y(\xi, \eta)$ values.
 - Plot the results to confirm the mapping.

Example

$$x(\xi, \eta) = 0.5 \cdot \xi\eta + 1.0 \cdot \xi + 0.5 \cdot \eta$$

$$y(\xi, \eta) = 0.5 \cdot \xi\eta + 0.5 \cdot \xi + 1.0 \cdot \eta$$



What have we learned?

- A bilinear mapping is defined as:

$$x(\xi, \eta) = R_x \xi \eta + S_x \xi + T_x \eta + U_x$$

- The coefficients of an analytical mapping can be found using known/control point values.
- For the bilinear mapping example the transformation matrix is the same for $x(\xi, \eta)$ and $y(\xi, \eta)$.

$$\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} R_x \text{ or } R_y \\ S_x \text{ or } S_y \\ T_x \text{ or } T_y \\ U_x \text{ or } U_y \end{pmatrix} = \begin{pmatrix} a_x \text{ or } a_y \\ b_x \text{ or } b_y \\ c_x \text{ or } c_y \\ d_x \text{ or } d_y \end{pmatrix}$$

- Once the mapping coefficients are found, the mapping is defined and any (ξ, η) can be mapped from the computational domain to the physical domain, (x, y) .